

Question	Answer	Marks
4(a)(i)	$p = mv$	C1
	$= 0.2(00) \times 6.(00) \times \sin 60(.0)^{\circ}$ or $0.2(00) \times 6.(00) \times \cos 30(.0)^{\circ}$	A1
	$= 1.04 \text{ kg m s}^{-1}$	
4(a)(ii)	$0.300 \times v_x \times \sin 60.0^{\circ} = 1.04$ $v_x = 4.00 \text{ m s}^{-1}$	A1
4(a)(iii)	$0.30 \times 4.0 \times \cos 60^{\circ}$ or $0.20 \times 6.0 \times \cos 60^{\circ}$ or $(0.30 + 0.20)v$ or $0.50v$	C1
	$0.30 \times 4.0 \times \cos 60^{\circ} + 0.20 \times 6.0 \times \cos 60^{\circ} = (0.30 + 0.20)v$ or $0.50v$	A1
	so $v = 2.4 \text{ m s}^{-1}$	
4(b)(i)	$E = \frac{1}{2}mv^2$	C1
	$\frac{1}{2} \times 0.50 \times 2.4^2 = \frac{1}{2} \times 72 \times x^2$	C1
	$x = 0.20 \text{ m}$	A1
4(b)(ii)	1. straight line from the origin sloping upwards	B1
	2. line drawn from a positive value of E_k at $x = 0$ to a positive value of x at $E_k = 0$	M1
	line has an increasing downwards slope	A1